

Consolidated Wellbeing Scoring Definition

One Earth Balance Sheet — Malaysia 2026

Version: 3.0 (Consolidated & Fixed) by Yagyaansh

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Status: MASTER REFERENCE — supersedes IMPLEMENTATION_SPECIFICATION.md and FRAMEWORK_EXTENSION_ANALYSIS.md

Purpose: Single source of truth for all wellbeing scoring logic, data requirements, and mapping to app questions

How This Document Is Organized

1. **Part A** — Individual Wellbeing Scoring (7 spheres, 61 indicators)
2. **Part B** — Village Resource Scoring (11 sub-domains)
3. **Part C** — Integration Formula (Individual + Village → Sustainability Score)
4. **Part D** — App Question → Use Case Mapping (what each of the 445 questions is for)
5. **Part E** — Gaps & Missing Questions (what the app needs but doesn't collect yet)

SCORING PRINCIPLES

All scores are **0 to 1** (normalized).

Individual Sphere Score = simple average of all valid indicator scores within that sphere

Individual Overall Score = simple average of 7 sphere scores (equal weight)

Village Sphere Score = simple average of all individual sphere scores in the village

Village Overall Score = simple average of 7 village sphere scores

Aggregation: Always use **arithmetic mean** (simple average). No multiplication.

- *Rationale:* A simple average ensures that a single deficit does not disproportionately wipe out the rest of a person's wellbeing score. If an individual has a perfectly stable income (1.0) but currently lacks liquid savings (0.0), a multiplicative formula ($1.0 * 0.0$) would erroneously state they have zero occupational wellbeing. An average (0.5) accurately reflects that their basic material resilience is intact despite lacking a buffer.
- FIX from Bhutan pilot: Occupational sphere (then called Financial) used `income * savings`; changed to averaging.

Missing Data:

- NA (Not Applicable) or NC (Not Collected): exclude from average (reduce denominator)
- DK (Don't Know) or RF (Refused): score as 0.5 (conservative neutral)
- Minimum for a valid sphere score: at least 1 indicator must have data

Minimum Village Sample: 30% of population OR 10 individuals, whichever is higher.

PART A: INDIVIDUAL WELLBEING SCORING

7 Spheres × 3 Tiers = 61 Total Indicators

Sphere	Tier 1	Tier 2	Tier 3	Total
Physical	1	5	4	10
Occupational	2	6	2	10
Emotional	2	4	1	7
Social	2	5	1	8
Environmental	2	6	2	10
Intellectual	3	3	2	8
Spiritual	2	5	1	8
TOTAL	14	34	13	61

Tier 1 = Bhutan pilot (proven, immediate deployment)

Tier 2 = Malaysia 2026 extension (survey-based, high impact)

Tier 3 = Future (requires equipment or clinical evaluation)

Tier 1 + Tier 2 operational set = 48 indicators (14 + 34)

SPHERE 1: PHYSICAL WELLBEING

P1.1: Body Mass Index (BMI) — Tier 1

App Questions: Q0206 (Height), Q0207 (Weight) — already in Demographic Information

Formula:

$BMI = weight_kg / (height_m ** 2)$

FIXED: Graduated scoring (was binary in Bhutan pilot)

if 18.5 <= BMI <= 24.9:

 score = 1.0 # Normal

elif 25.0 <= BMI <= 29.9:

 score = 0.6 # Overweight

elif 17.0 <= BMI < 18.5:

 score = 0.6 # Mildly underweight

elif 30.0 <= BMI <= 34.9:

 score = 0.3 # Obese Class I

elif 15.0 <= BMI < 17.0:

```

        score = 0.3      # Moderately underweight

else:

    score = 0.0      # Severe obesity or severe underweight

```

Fix Applied: Bhutan pilot used binary (1 if normal, 0 otherwise). Now graduated to avoid penalizing borderline cases.

P2.1: Sleep Quality & Duration — Tier 2

App Questions: NOT IN APP — needs to be added

Wellbeing Survey Q15a/Q15b

Formula:

```

# Duration

if sleep_hours < 5:      duration = 0

elif sleep_hours < 6:    duration = 0.3

elif sleep_hours < 7:    duration = 0.6

elif sleep_hours <= 9:   duration = 1.0

else:                   duration = 0.6 # oversleeping


# Quality

if quality == "Good":    quality_score = 1.0

elif quality == "Fair":  quality_score = 0.5

else:                   quality_score = 0.0


score = (duration + quality_score) / 2

```

P2.2: Physical Activity — Tier 2

App Questions: NOT IN APP — needs to be added

Wellbeing Survey Q16a/Q16b

Formula:

```

# Frequency

```

```

if freq == "5+ days/week":    freq_score = 1.0

elif freq == "3-4 days/week": freq_score = 0.7

elif freq == "1-2 days/week": freq_score = 0.4

else:                          freq_score = 0.0

```

```

# Duration (only if exercises)

```

```

if duration == ">40 min":      dur_score = 1.0

elif duration == "20-40 min": dur_score = 0.75

elif duration == "<20 min":    dur_score = 0.5

else:                          dur_score = None

```

```

if freq_score == 0:

```

```

    score = 0

```

```

else:

```

```

    score = (freq_score + dur_score) / 2

```

P2.3: Dietary Balance — Tier 2

App Questions: Q0205 (Diet type — partial), Q0131-Q0202 (Food Consumption — can derive dietary diversity)

Wellbeing Survey Q17a/Q17b

Formula:

```

# Balance

```

```

if balance == "Good":          balance_score = 1.0

elif balance == "Fair":        balance_score = 0.5

else:                          balance_score = 0.0

```

```

# Meal regularity

```

```

if regularity == "Regular":      reg_score = 1.0

elif regularity == "Somewhat":   reg_score = 0.5

else:                            reg_score = 0.0

```

```

score = (balance_score + reg_score) / 2

```

Note: Can also be computed from Food Consumption data (Q0131-Q0202) by counting distinct food categories consumed. If ≥ 9 categories = 1.0, 6-8 = 0.6, 3-5 = 0.3, < 3 = 0.

P2.4: Energy Levels — Tier 2

App Questions: NOT IN APP — needs to be added

Wellbeing Survey Q18

Formula:

```

if energy == "High/Upbeat":  score = 1.0

elif energy == "Moderate":   score = 0.5

else:                        score = 0.0

```

P2.5: Dental Hygiene — Tier 2

App Questions: NOT IN APP — needs to be added

Wellbeing Survey Q19a/Q19b

Formula:

Brushing

```

if brushing == "Twice+ daily": brush = 1.0

elif brushing == "Once daily":  brush = 0.5

else:                           brush = 0.0

```

Problems

```

if problems == "None":         prob = 1.0

elif problems == "Occasionally": prob = 0.5

else:                          prob = 0.0

```

```

score = (brush + prob) / 2

```

P3.1: Blood Pressure — Tier 3 (requires equipment)

App Questions: NOT IN APP — requires health measurement module

Wellbeing Survey Q20a/Q20b

Question format:

Q20a: Systolic blood pressure (mmHg): _____

Q20b: Diastolic blood pressure (mmHg): _____

Validation:

- Acceptable device range: systolic 70-260, diastolic 40-150
- Measure after 5 minutes seated rest
- If first reading is high, repeat and average two readings

Formula:

```
if systolic < 120 and diastolic < 80:

    score = 1.0

elif 120 <= systolic <= 129 and diastolic < 80:

    score = 0.7

elif (130 <= systolic <= 139) or (80 <= diastolic <= 89):

    score = 0.4

elif (140 <= systolic <= 179) or (90 <= diastolic <= 119):

    score = 0.1

else: # systolic >= 180 or diastolic >= 120

    score = 0.0
```

P3.2: Blood Sugar — Tier 3 (requires testing)

App Questions: NOT IN APP — requires health measurement module

Wellbeing Survey Q21a/Q21b

Question format:

Q21a: Fasting blood glucose (mg/dL): _____

Q21b: If fasting unavailable, random blood glucose (mg/dL): _____

Validation:

- Acceptable range: 40-500 mg/dL
- Prefer fasting value; use random value only when fasting is unavailable

Formula:

```
if fasting_glucose is not None:
```

```
    if fasting_glucose < 100:
```

```
        score = 1.0
```

```
    elif fasting_glucose <= 125:
```

```
        score = 0.5
```

```
    else: # >= 126
```

```
        score = 0.0
```

```
else:
```

```
    # fallback to random glucose
```

```
    if random_glucose < 140:
```

```
        score = 1.0
```

```
    elif random_glucose <= 199:
```

```
        score = 0.5
```

```
    else: # >= 200
```

```
        score = 0.0
```

P3.3: Cholesterol — Tier 3 (requires blood test)

App Questions: NOT IN APP — requires lab/testing module

Wellbeing Survey Q22**Question format:**

Q22: Total cholesterol (mg/dL): _____

Validation:

- Acceptable range: 80-500 mg/dL

- Use latest test from last 12 months if fresh test unavailable

Formula:

```
if total_cholesterol < 200:

    score = 1.0

elif total_cholesterol <= 239:

    score = 0.5

else: # >= 240

    score = 0.0
```

P3.4: Chronic Disease History — Tier 3

App Questions: Q0216 (Chronic Diseases) — partial coverage
Wellbeing Survey Q23a/Q23b

Question format:

Q23a: Do you have any chronic condition diagnosed by a health professional?

Q23b: If yes, is it currently managed and under control?

Validation:

- Use only diagnosed conditions (not self-diagnosed)
- Managed = treatment adherence and no recent acute event

Formula:

```
if chronic_diseases == [] or chronic_diseases == ["None"]:

    score = 1.0

elif all_diagnosed_conditions_managed:

    score = 0.5

else: # one or more unmanaged chronic conditions

    score = 0.0
```

Physical Sphere Score:

Tier 1 only

Physical_T1 = BMI_score

Tier 1+2 (Malaysia 2026 target)

Physical_T1T2 = average([BMI, Sleep, Exercise, Diet, Energy, Dental])

Tier 1+2+3 (future)

Physical_full = average([BMI, Sleep, Exercise, Diet, Energy, Dental,
BloodPressure, BloodSugar, Cholesterol, ChronicDisease])

SPHERE 2: OCCUPATIONAL WELLBEING

Definition Note: For Malaysia 2026, the sphere formerly labeled Financial Wellbeing is renamed Occupational Wellbeing to align with the Masters of Life framing. In the current pilot, Occupational Wellbeing is operationalized through livelihood stability, income adequacy, savings, debt resilience, asset ownership, and financial or enterprise literacy. Broader work-quality and vocation-fit indicators can be added in later iterations.

O1.1: Annual Livelihood Income Level - Tier 1

App Questions: Q0212 (Yearly Income per member) - already in Demographic Information

Formula:

Thresholds must be set per country (bio-region contextualization)

Example: Malaysia RM 20,000 low / RM 50,000 high

```
if income == "No income":           score = 0.0
elif income < THRESHOLD_LOW:         score = 0.33
elif income <= THRESHOLD_HIGH:       score = 0.67
else:                                score = 1.0
```

O1.2: Savings & Investment Status - Tier 1

App Questions: Q0214 (Savings/Investments Y/N), Q0215 (Amount) - already in Demographics

Formula:

```
# FIXED: "No savings" now scores 0 instead of 0.5

# Rationale: 0.5 for "no savings" inflated occupational wellbeing scores unfairly

if savings == "Yes":      score = 1.0

elif savings == "No":     score = 0.0

else:                     score = RF # refused
```

Fix Applied: Bhutan pilot scored "No savings" as 0.5. Changed to 0.0 for accuracy.

O2.1: Livelihood Adequacy (Normalized) - Tier 2

App Questions: NOT IN APP - needs to be added (subjective perception)

Wellbeing Survey Q26

Formula:

```
if adequacy == "Abundant":      score = 1.0

elif adequacy == "Comfortable":  score = 0.8

elif adequacy == "Sufficient":   score = 0.6

elif adequacy == "Barely sufficient": score = 0.3

else:                           score = 0.0 # Insufficient
```

O2.2: Occupational Security Feeling - Tier 2

App Questions: NOT IN APP - needs to be added

Wellbeing Survey Q27

Formula:

```
if security == "Secure":        score = 1.0

elif security == "Somewhat secure": score = 0.5

else:                           score = 0.0
```

O2.3: Emergency Fund - Tier 2

App Questions: NOT IN APP - needs to be added

Wellbeing Survey Q28

Formula:

```
if fund == "6+ months":      score = 1.0

elif fund == "3-6 months":   score = 0.7

elif fund == "1-3 months":   score = 0.4

elif fund == "<1 month":      score = 0.1

else:                         score = 0.0 # No savings
```

O2.4: Debt Burden - Tier 2

App Questions: NOT IN APP - needs to be added

Wellbeing Survey Q29

Formula:

```
if debt == "No debt":        score = 1.0

elif debt == "Manageable":   score = 0.6

elif debt == "Heavy":        score = 0.3

else:                        score = 0.0 # Overwhelming
```

O2.5: Asset Ownership - Tier 2

App Questions: DERIVABLE from Q0266 (land count), Q0289 (house count), Q0351 (vehicle count), Q0062 (livestock types), Q0399 (business Y/N)

Formula:

```
has_land = 1 if Q0266 > 0 else 0

has_house = 1 if Q0289 > 0 else 0

has_vehicle = 1 if Q0351 > 0 else 0

has_livestock = 1 if Q0061 > 0 else 0

has_business = 1 if Q0399 == "Yes" else 0

asset_count = has_land + has_house + has_vehicle + has_livestock + has_business
```

```

if asset_count == 0:      score = 0.0

elif asset_count <= 2:    score = 0.4

elif asset_count <= 4:    score = 0.7

else:                    score = 1.0

```

Note: This can be computed entirely from existing app data - no new question needed.

O2.6: Financial & Enterprise Literacy - Tier 2

App Questions: Q0213 (Bank Account Y/N) - partial; Q0228 (Entrepreneurial & Business Skills including Financial Literacy) - partial

Wellbeing Survey Q31a/Q31b

Formula:

```

if budget_tracking == "Yes":      budget = 0.5

else:                             budget = 0.0


if financial_concepts == "Yes, well": concepts = 0.5

elif financial_concepts == "Somewhat": concepts = 0.25

else:                             concepts = 0.0


score = budget + concepts  # 0 to 1

```

O3.1: Credit Score - Tier 3 (if accessible)

App Questions: NOT IN APP - requires financial records module

Wellbeing Survey Q32a

Question format:

Q32a: If available, what is your latest credit score?

Validation:

- Accept score only from recognized bureau/bank report
- If country/region has no credit scoring system, mark as NA

Formula:

```
if credit_score is None:

    score = NA

elif credit_score >= 750:

    score = 1.0

elif credit_score >= 700:

    score = 0.8

elif credit_score >= 650:

    score = 0.6

elif credit_score >= 600:

    score = 0.3

else:

    score = 0.1
```

O3.2: Livelihood Diversification - Tier 3

App Questions: DERIVABLE from existing income sections (farm, business, employment, tourism, remittance)

Wellbeing Survey Q32b**Question format:**

Q32b: How many distinct livelihood or income sources does your household currently have?

Validation:

- Count only active income sources in last 12 months
- Exclude one-off windfalls

Formula:

```
# Sources: farming, salaried work, business, tourism/services, remittance/pension, other

if source_count >= 4:

    score = 1.0
```

```

elif source_count == 3:

    score = 0.75

elif source_count == 2:

    score = 0.5

elif source_count == 1:

    score = 0.25

else:

    score = 0.0

```

Occupational Sphere Score:

```

# FIXED: Use averaging, NOT multiplication

# Bhutan pilot used: income_score * savings_score - this is WRONG

# A person with high income but no savings would score 0 (too harsh)


# Tier 1 only

Occupational_T1 = average([income_score, savings_score])


# Tier 1+2 (Malaysia 2026 target)

Occupational_T1T2 = average([income, savings, adequacy, security,

                             emergency_fund, debt, assets, fin_literacy])


# Aggregate Score: Village material needs, Individual and Village Income, Income Adequacy

# Tier 1+2+3 (future)

Occupational_full = average([income, savings, adequacy, security,

                             emergency_fund, debt, assets, fin_literacy,

                             credit_score, income_diversification, material_needs_met_pct])

```

Fix Applied: Changed from multiplication to averaging for consistency.

SPHERE 3: EMOTIONAL WELLBEING

E1.1: Current Emotional State — Tier 1

App Questions: Q0218 (Overall Well-being) — already in Demographics > Mental & Emotional Well-being

Formula:

```
if state == "Mentally and Emotionally Fit": score = 1.0

elif state == "Generally Stable":           score = 0.5

elif state == "Facing Challenges":         score = 0.25

else:                                       score = RF # prefer not to answer
```

Note: App has 4 options including "Facing Challenges" (not in Bhutan pilot). Score adjusted.

E1.2: Support System Availability — Tier 1

App Questions: Q0220 (Support System) — already in Demographics

Formula:

```
# App allows multi-select: Family/Friends, Professional, Social Media, Personal Coping, Prefer not to
support_count = len(selected_options) # excluding "Prefer not to answer"

if "Prefer not to answer" in selection:

    score = RF

elif support_count >= 2:

    score = 1.0

elif support_count == 1:

    score = 0.5

else:

    score = 0.0
```

Note: App provides richer data than Bhutan pilot. Multiple support types = stronger support.

E2.1: Stress Frequency — Tier 2

App Questions: Q0221 (Stress and Resilience) — already in Demographics

Formula:

```
# App options: "Low Stress, High Resilience" | "Moderate Stress, Moderate Resilience" |  
  
# "High Stress, Low Resilience" | "Prefer not to answer"  
  
if stress == "Low Stress, High Resilience":      score = 1.0  
  
elif stress == "Moderate Stress, Moderate Resilience": score = 0.5  
  
elif stress == "High Stress, Low Resilience":      score = 0.0  
  
else:                                              score = RF
```

E2.2: Anger/Temperament Management — Tier 2

App Questions: NOT IN APP — needs to be added

Wellbeing Survey Q33

Formula:

```
if anger == "Rarely":      score = 1.0  
  
elif anger == "Occasionally": score = 0.5  
  
else:                      score = 0.0
```

E2.3: Connection Quality — Tier 2

App Questions: Q0219 (Communication of Feelings) — partial proxy

Wellbeing Survey Q34

Formula:

```
if connection == "Deeply connected":  score = 1.0  
  
elif connection == "Somewhat connected": score = 0.5  
  
else:                                score = 0.0
```

E2.4: Life Satisfaction — Tier 2

App Questions: NOT IN APP — needs to be added

Wellbeing Survey Q35

Formula:

```

if satisfaction == "Very satisfied":  score = 1.0

elif satisfaction == "Satisfied":      score = 0.7

elif satisfaction == "Neutral":        score = 0.5

elif satisfaction == "Unsatisfied":    score = 0.2

else:                                  score = 0.0

```

E3.1: Mental Health Professional Assessment — Tier 3

App Questions: NOT IN APP — requires optional clinical partner pathway

Wellbeing Survey Q36

Question format:

Q36: If assessed by a qualified mental health professional in the last 12 months,
what was the overall assessment category?

Validation:

- Accept only licensed professional assessment
- If no assessment available, mark as NA (do not force score)

Formula:

```

if assessment == "Healthy / no significant concern":

    score = 1.0

elif assessment == "Mild concern":

    score = 0.7

elif assessment == "Moderate concern":

    score = 0.4

elif assessment == "Severe concern":

    score = 0.1

else:  # crisis / urgent

    score = 0.0

```

Emotional Sphere Score:

```
Emotional_T1 = average([state_score, support_score])

Emotional_T1T2 = average([state, support, stress, anger, connection, satisfaction])

Emotional_full = average([state, support, stress, anger, connection, satisfaction,
                           mental_health_assessment])
```

SPHERE 4: SOCIAL WELLBEING

S1.1: Cultural Participation — Tier 1

App Questions: Q0231 (Cultural and Traditional Hobbies) — partially covers this

Formula:

```
# App options include: Performing Traditional Dance, Storytelling,
# Traditional Crafts, Participating in Festivals/Rituals

if len(Q0231_selections) > 0:  score = 1.0

else:                          score = 0.0
```

S1.2: Empathy Level — Tier 1

App Questions: Q0222 (Empathy) — already in Demographics

Formula:

```
if empathy == "High":          score = 1.0

elif empathy == "Moderate":    score = 0.5

elif empathy == "Low":         score = 0.25

elif empathy == "None":        score = 0.0

else:                          score = RF
```

S2.1: Festival/Community Event Participation — Tier 2

App Questions: Q0231 (includes "Participating in Festivals/Rituals") — partial

Formula:

```
if participation == "Active/Regular": score = 1.0

elif participation == "Rare/Sometimes": score = 0.5

else:                                score = 0.0
```

Note: App only captures whether they participate, not frequency. May need enhancement.

S2.2: Neighbor Relations — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if relations == "Excellent":  score = 1.0
elif relations == "Good":      score = 0.75
elif relations == "Neutral":    score = 0.5
elif relations == "Poor":       score = 0.25
else:                           score = 0.0  # Conflict
```

S2.3: Community Contribution — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if contribution == "Active":    score = 1.0
elif contribution == "Occasional": score = 0.6
elif contribution == "Minimal": score = 0.3
else:                           score = 0.0
```

S2.4: Sense of Belonging — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if belonging == "Strong":       score = 1.0
elif belonging == "Moderate":    score = 0.6
elif belonging == "Weak":        score = 0.3
else:                           score = 0.0
```

S2.5: Social Conflicts (Past Year) — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if conflicts == "None":         score = 1.0
elif conflicts == "Minor":      score = 0.6
elif conflicts == "Moderate":   score = 0.3
else:                           score = 0.0  # Severe
```

S3.1: Social Network Size — Tier 3

App Questions: NOT IN APP — needs to be added

Wellbeing Survey Q44

Question format:

Q44: How many close relationships do you have that you can rely on in times of need?

Validation:

- "Close relationship" means person is trusted and available for meaningful support
- Exclude purely transactional contacts

Formula:

```
if close_relationships >= 5:

    score = 1.0

elif close_relationships >= 3:

    score = 0.7

elif close_relationships >= 1:

    score = 0.4

else:

    score = 0.0
```

Social Sphere Score:

```
Social_T1 = average([cultural_score, empathy_score])

Social_T1T2 = average([cultural, empathy, festivals, neighbors,

                      contribution, belonging, conflicts])

Social_full = average([cultural, empathy, festivals, neighbors,

                      contribution, belonging, conflicts, social_network_size])
```

SPHERE 5: ENVIRONMENTAL WELLBEING

EN1.1: Water Quality — Drinking/Cooking — Tier 1

App Questions: Q0315 (Water Quality for Cooking & Drinking) — already in Water section

Formula:

```
if quality == "Potable":          score = 1.0

elif quality == "Needs Boiling/Filtration": score = 0.5

elif quality == "Slightly Contaminated":  score = 0.25

else:                                   score = 0.0 # Heavily contaminated
```

Note: App has 4 options (more granular than Bhutan pilot's 3). Score adjusted.

EN1.2: Water Quality — Sanitation/Bathing — Tier 1

App Questions: Q0319 (Water Quality for Sanitation & Bathing) — already in Water section

Formula:

```
if quality == "Soft Water":      score = 1.0

elif quality == "Hard Water":    score = 0.5

else:                            score = 0.0 # Contaminated
```

EN2.1: Air Quality (Perceived) — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if air == "Clean and fresh":  score = 1.0

elif air == "Moderate":      score = 0.5

else:                        score = 0.0
```

EN2.2: Green Space Access — Tier 2

App Questions: Q0362 (Total Area Under Forest Cover) — can partially derive

Formula:

```
# Subjective perception

if access == "Yes, regularly":  access_score = 1.0
```

```
elif access == "Limited":      access_score = 0.5

else:                          access_score = 0.0
```

Distance

```
if distance == "<5 min walk":   dist = 1.0

elif distance == "5-15 min":    dist = 0.7

elif distance == "15-30 min":   dist = 0.4

else:                           dist = 0.0
```

```
if access_score == 0: score = 0

else: score = (access_score + dist) / 2
```

EN2.3: Neighborhood Cleanliness — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if clean == "Clean and well-maintained": score = 1.0

elif clean == "Moderately clean":        score = 0.5

else:                                     score = 0.0
```

EN2.4: Garbage Management — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if garbage == "Proper organized system": score = 1.0

elif garbage == "Informal system":       score = 0.5

else:                                     score = 0.0
```

EN2.5: Land/Space Availability — Tier 2

App Questions: DERIVABLE from Q0266 (Land Count), Q0268 (Total Area), Q0203 (Family Size)

Formula:

```
# Can compute: total land area / family members
```

```
# OR use subjective question
```

```
if space == "Ample":      score = 1.0
```

```
elif space == "Adequate": score = 0.6
```

```
elif space == "Cramped":  score = 0.3
```

```
else:                      score = 0.0
```

EN2.6: Food Quality Access — Tier 2

App Questions: DERIVABLE from Food Consumption section (Q0131-Q0202) — ratio of self-grown to purchased

Formula:

```
# Option 1: Subjective
```

```
if food_quality == "Fresh/organic readily available": score = 1.0
```

```
elif food_quality == "Mix of fresh and processed":    score = 0.5
```

```
else:                                                  score = 0.0
```

```
# Option 2: Computed from app data
```

```
# self_grown_ratio = sum(self_grown) / sum(total_consumed) across all food types
```

```
# score = self_grown_ratio (0 to 1, higher = more self-sufficient)
```

EN3.1: Air Quality Index (AQI) — Tier 3 (requires monitoring equipment)

App Questions: NOT IN APP — requires AQI measurement entry

Wellbeing Survey Q53a

Question format:

Q53a: Average local AQI reading (last 7 days): _____

Validation:

- AQI range should be 0-500
- Prefer nearest official station; fallback to calibrated field monitor

Formula:

```
if aqi <= 50:

    score = 1.0

elif aqi <= 100:

    score = 0.7

elif aqi <= 150:

    score = 0.4

elif aqi <= 200:

    score = 0.2

else:

    score = 0.0
```

EN3.2: Water TDS/pH Testing — Tier 3 (requires test kit)

App Questions: NOT IN APP — requires test module

Wellbeing Survey Q53b/Q53c**Question format:**

Q53b: Drinking water TDS (ppm): _____

Q53c: Drinking water pH: _____

Validation:

- TDS range: 0-3000 ppm
- pH range: 0-14
- Test at point-of-use, not only at source

Formula:

```
# TDS score
```

```

if tds <= 300:

    tds_score = 1.0

elif tds <= 500:

    tds_score = 0.7

elif tds <= 1000:

    tds_score = 0.3

else:

    tds_score = 0.0


# pH score

if 6.5 <= ph <= 8.5:

    ph_score = 1.0

elif (6.0 <= ph < 6.5) or (8.5 < ph <= 9.0):

    ph_score = 0.6

else:

    ph_score = 0.0


score = average([tds_score, ph_score])

```

Environmental Sphere Score:

```

Environmental_T1 = average([drinking_water, sanitation_water])

Environmental_T1T2 = average([drinking_water, sanitation_water, air,

                               green_space, cleanliness, garbage,

                               land_space, food_quality])

Environmental_full = average([drinking_water, sanitation_water, air,

                               green_space, cleanliness, garbage,

                               land_space, food_quality, aqi_score, water_test_score])

```

SPHERE 6: INTELLECTUAL WELLBEING

I1.1: Highest Education Level — Tier 1

App Questions: Q0224 (Current Education Status) — already in Demographics

Formula:

```
# App has more options than Bhutan pilot. Mapping:
```

```
higher = ["Diploma", "Bachelor's", "Master's", "Doctorate"]
```

```
secondary = ["Higher Secondary", "Secondary"]
```

```
primary = ["Primary"]
```

```
other = ["Monastic Education", "Vocational/Skill-based Training", "Religious Education"]
```

```
if education in higher:      score = 1.0
```

```
elif education in secondary: score = 0.6
```

```
elif education in other:     score = 0.5 # non-formal education has value
```

```
elif education in primary:   score = 0.3
```

```
else:                        score = 0.0 # No Formal Education
```

I1.2: Currently Seeking Education — Tier 1

App Questions: Q0225 (Education Seeking to Gain) — already in Demographics

Formula:

```
if Q0225 has any selection (not empty/None):
```

```
    score = 1.0
```

```
else:
```

```
    score = 0.0
```

I1.3: Seeking Technical/Vocational Skills — Tier 1

App Questions: Q0236-Q0242 (Skills Seeking to Learn) — already in Demographics

Formula:

```
if any of Q0236 through Q0242 has selections:
```

```
    score = 1.0
```

```
else:
```

```
    score = 0.0
```

I2.1: Value of Education/Learning — Tier 2

App Questions: Q0244 (Educational Aspirations) — partial proxy

Formula:

```
if value == "Very important":    score = 1.0
elif value == "Somewhat important": score = 0.5
else:                             score = 0.0
```

I2.2: Learning Engagement — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if engagement == "Regularly":    score = 1.0
elif engagement == "Occasionally": score = 0.5
elif engagement == "Rarely":     score = 0.25
else:                             score = 0.0
```

I2.3: Information Access — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if access == "Yes, regular":    score = 1.0
elif access == "Limited":      score = 0.5
else:                           score = 0.0
```

I3.1: Critical Thinking Assessment — Tier 3 (requires test)

App Questions: NOT IN APP — requires mini-assessment module

Wellbeing Survey Q57a

Question format:

Q57a: 10-question critical reasoning mini-test (score out of 10)

Validation:

- Standardized question bank with rotating variants
- Timed (10-15 minutes), invigilated by enumerator

Formula:

```
score = max(0.0, min(1.0, correct_answers / 10.0))
```

I3.2: Creativity/Innovation — Tier 3

App Questions: NOT IN APP — needs to be added

Wellbeing Survey Q57b

Question format:

Q57b: How often do you engage in creative or innovative activities

(problem-solving, designing, creating, inventing)?

Validation:

- Activities should be concrete examples from last 6 months
- Enumerator records one example for verification

Formula:

```
if creativity == "Weekly or more":  
  
    score = 1.0  
  
elif creativity == "Monthly":  
  
    score = 0.7  
  
elif creativity == "A few times per year":  
  
    score = 0.4  
  
else:  
  
    score = 0.0
```

Intellectual Sphere Score:

```
Intellectual_T1 = average([education, seeking_edu, seeking_skills])  
  
Intellectual_T1T2 = average([education, seeking_edu, seeking_skills,  
                             edu_value, learning_engagement, info_access])  
  
Intellectual_full = average([education, seeking_edu, seeking_skills,  
                             edu_value, learning_engagement, info_access,  
                             critical_thinking, creativity_innovation])
```

SPHERE 7: SPIRITUAL WELLBEING

SP1.1: Spiritual Self-Evaluation — Tier 1

App Questions: NOT IN APP — needs to be added

Formula:

```
if spirituality == "Highly spiritual":      score = 1.0
elif spirituality == "Moderately spiritual": score = 0.5
elif spirituality == "Somewhat spiritual":   score = 0.25
else:                                       score = 0.0
```

SP1.2: Ethical/Moral Consideration — Tier 1

App Questions: NOT IN APP — needs to be added

Note: Bhutan pilot used "Consider Karma in Daily Life" — generalized to ethical/moral principles

Formula:

```
if ethics == "Regularly":      score = 1.0
elif ethics == "Occasionally":  score = 0.5
else:                          score = 0.0
```

SP2.1: Meditation/Prayer Practice — Tier 2

App Questions: Q0251 (Spiritual Aspirations) mentions "Meditation/Mindfulness" but as aspiration, not practice

Formula:

```
if meditation == "Daily":      score = 1.0
elif meditation == "Weekly":    score = 0.7
elif meditation == "Monthly":   score = 0.4
else:                          score = 0.0
```

SP2.2: Time in Silence/Reflection — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if reflection == "Daily":      score = 1.0
elif reflection == "Weekly":    score = 0.6
elif reflection == "Rarely":    score = 0.3
else:                          score = 0.0
```

SP2.3: Sense of Purpose — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if purpose == "Strong":      score = 1.0

elif purpose == "Moderate":  score = 0.6

elif purpose == "Weak":      score = 0.3

else:                        score = 0.0
```

SP2.4: Gratitude Practice — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if gratitude == "Regularly":  score = 1.0

elif gratitude == "Occasionally": score = 0.5

elif gratitude == "Rarely":   score = 0.25

else:                         score = 0.0
```

SP2.5: Mindfulness in Daily Actions — Tier 2

App Questions: NOT IN APP — needs to be added

Formula:

```
if mindfulness == "Very mindful":  score = 1.0

elif mindfulness == "Somewhat mindful": score = 0.6

else:                               score = 0.3
```

SP3.1: Journaling/Self-Reflection — Tier 3

App Questions: NOT IN APP — needs to be added

Wellbeing Survey Q65

Question format:

Q65: How often do you journal or intentionally reflect on your life and actions?

Validation:

- Reflection can be written, audio, or structured personal reflection practice
- Must be recurring behavior (not one-off)

Formula:

```

if journaling == "Daily":

    score = 1.0

elif journaling == "Weekly":

    score = 0.7

elif journaling == "Monthly":

    score = 0.4

elif journaling == "Rarely":

    score = 0.2

else:

    score = 0.0

```

Spiritual Sphere Score:

```

Spiritual_T1 = average([spirituality, ethics])

Spiritual_T1T2 = average([spirituality, ethics, meditation, reflection,

                           purpose, gratitude, mindfulness])

Spiritual_full = average([spirituality, ethics, meditation, reflection,

                           purpose, gratitude, mindfulness, journaling_reflection])

```

OVERALL INDIVIDUAL WELLBEING SCORE

```

Individual_Wellbeing = average([

    Physical_score,

    Occupational_score,

    Emotional_score,

    Social_score,

```

```

    Environmental_score,

    Intellectual_score,

    Spiritual_score

])

# Range: 0 to 1

# Interpretation: 0-0.3 Low, 0.3-0.6 Moderate, 0.6-0.8 Good, 0.8-1.0 Excellent

```

PART B: VILLAGE RESOURCE SCORING (11 Sub-Domains)

Village resource scores are computed from aggregated household data and village-level measurements. These feed into the **Balance Sheet** (Goal 5) and the **Village Resource Adequacy** component of the overall sustainability score.

Domain 1: Natural Resources & Environment

Sub-Domain 1: LAND

App Questions: Q0266-Q0288 (Landholding & Usage)

Balance Sheet: Land Balance Sheet

Key Metrics & Scoring:

```

# Population Density

total_village_area_km2 = sum(all_household_land_areas) # approximate

population_density = village_population / total_village_area_km2


if 400 <= density <= 2000:      density_score = 1.0

elif 2000 < density <= 5000:    density_score = linear(2000, 1.0, 5000, 0.5)

elif density > 5000:           density_score = linear(5000, 0.5, 10000, 0.0)

else:                          density_score = 0.8 # underpopulated


# Forest Cover (requires village-level data + Q0362)

forest_ratio = total_forest_area / total_village_area

```

```

if forest_ratio >= 0.60:          forest_score = 1.0

elif forest_ratio >= 0.40:        forest_score = linear(0.40, 0.5, 0.60, 1.0)

else:                             forest_score = linear(0.0, 0.0, 0.40, 0.5)

```

```

# Land Utilization

```

```

utilized_land = sum(lands where Q0271 == "Yes")

total_owned_land = sum(Q0268 for all households)

utilization_ratio = utilized_land / total_owned_land

utilization_score = utilization_ratio # 0 to 1

```

```

Land_Score = average([density_score, forest_score, utilization_score])

```

Sub-Domain 2: WATER

App Questions: Q0313-Q0337 (Water)

Balance Sheet: Water Balance Sheet

Key Metrics & Scoring:

```

# Water quality (aggregated from household data)

drinking_quality_scores = [score(Q0315) for each household]

village_drinking_score = average(drinking_quality_scores)

```

```

# Water scarcity

```

```

scarcity_count = count(Q0335 == "Yes")

scarcity_ratio = scarcity_count / total_households

if scarcity_ratio == 0:          scarcity_score = 1.0

elif scarcity_ratio < 0.2:        scarcity_score = 0.7

elif scarcity_ratio < 0.5:        scarcity_score = 0.4

else:                             scarcity_score = 0.0

```

```
# Water harvesting adoption
```

```
harvesting_count = count(Q0330 has selections)
```

```
harvesting_ratio = harvesting_count / total_households
```

```
harvesting_score = harvesting_ratio
```

```
Water_Score = average([village_drinking_score, scarcity_score, harvesting_score])
```

Sub-Domain 3: FORESTRY & TIMBER

App Questions: Q0362-Q0374 (Forestry & Timber)

Balance Sheet: Forestry Balance Sheet

Key Metrics & Scoring:

```
# Forest cover (from Q0362 aggregated)
```

```
total_forest = sum(Q0362 for all households)
```

```
# Sustainable harvest (production versus requirements)
```

```
harvest = sum(Q0363)
```

```
requirement = sum(Q0370 where Q0369 == "Yes")
```

```
if requirement == 0: timber_balance = 1.0
```

```
else: timber_balance = min(1.0, harvest / requirement)
```

```
Forestry_Score = average([forest_score, timber_balance])
```

```
# forest_score reused from Land sub-domain
```

Sub-Domain 4: ENERGY & FUEL

App Questions: Q0338-Q0350 (Energy & Fuel)

Balance Sheet: Energy Balance Sheet

Key Metrics & Scoring:

```
# Grid connectivity
```

```

grid_ratio = count(Q0338 == "Yes") / total_households

grid_score = grid_ratio

# Renewable energy

renewable_sources = count(Q0349 selections)

if renewable_sources >= 3:    renewable_score = 1.0

elif renewable_sources >= 1: renewable_score = 0.5

else:                        renewable_score = 0.0

# Energy self-sufficiency (requires village-level renewable %)

# From Q0350: sum of all renewable % contributions

renewable_pct = sum(Q0350 values)

if renewable_pct >= 80:      self_sufficiency = 1.0

elif renewable_pct >= 50:    self_sufficiency = 0.7

elif renewable_pct >= 20:    self_sufficiency = 0.4

else:                        self_sufficiency = 0.1

Energy_Score = average([grid_score, renewable_score, self_sufficiency])

```

Domain 2: Human Settlements & Infrastructure

Sub-Domain 5: HOUSING

App Questions: Q0289-Q0312 (Housing)

Balance Sheet: Housing Balance Sheet

Key Metrics & Scoring:

```

# Space per person

total_built_area = sum(Q0294 for all houses)

total_population = sum(Q0203 for all households)

```

```

space_per_person = total_built_area / total_population

if space_per_person >= 25:      space_score = 1.0

elif space_per_person >= 15:    space_score = linear(15, 0.5, 25, 1.0)

else:                          space_score = linear(0, 0.0, 15, 0.5)


# Housing surplus/deficit

total_units = sum(Q0293)

total_households = count(households)

surplus_ratio = (total_units - total_households) / total_households

if surplus_ratio > 0.1:         surplus_score = 1.0

elif surplus_ratio >= -0.1:     surplus_score = 0.7

else:                          surplus_score = max(0, 0.7 + surplus_ratio * 3)


# Construction quality

# Score based on Q0299 (Construction Type)

# Traditional/Ecofriendly = 1.0, Wooden = 0.7, Concrete = 0.5, Other = 0.3

construction_scores = [score_construction(Q0299) for each house]

construction_avg = average(construction_scores)


Housing_Score = average([space_score, surplus_score, construction_avg])

```

Sub-Domain 6: COMMUNITY INFRASTRUCTURE

App Questions: No dedicated section in app — requires admin/external data feed

Balance Sheet: Community Infrastructure Balance Sheet

Data inputs needed:

- Population
- Education facilities (ECCD, primary, secondary)

- Health facilities (clinic count, pharmacy access)
- Civic facilities (post office, bank/financial point, town hall, security point)
- Environmental infra (% sewage connection, % waste-system coverage)
- Utilities (% streetlight coverage, % internet coverage, water-storage adequacy ratio)

Key Metrics & Scoring:

```

if population is None:

    Community_Infra_Score = None

else:

    # Education adequacy

    required_eccd = max(1, ceil(population / 500))

    required_primary = max(1, ceil(population / 1000))

    required_secondary = max(1, ceil(population / 2000))

    education_score = average([

        surplus_deficit_score(eccd_centers, required_eccd),

        surplus_deficit_score(primary_schools, required_primary),

        surplus_deficit_score(secondary_schools, required_secondary),

    ])

    # Health adequacy

    required_clinics = max(1, ceil(population / 2000))

    clinic_score = surplus_deficit_score(clinics, required_clinics)

    pharmacy_score = 1.0 if pharmacies_within_5km >= 1 else 0.0

    health_score = average([clinic_score, pharmacy_score])

    # Civic adequacy

    civic_score = average([

```

```

    1.0 if post_office_exists else 0.0,

    1.0 if bank_exists else 0.0,

    1.0 if town_hall_exists else 0.0,

    1.0 if security_post_exists else 0.0,

])

# Environmental infrastructure

sewage_score = sewage_connection_pct / 100.0

waste_score = waste_system_coverage_pct / 100.0

environmental_score = average([sewage_score, waste_score])

# Utilities

streetlight_score = street_light_coverage_pct / 100.0

internet_score = internet_coverage_pct / 100.0

storage_score = water_storage_adequacy_ratio # 0 to 1

utilities_score = average([streetlight_score, internet_score, storage_score])

Community_Infra_Score = average([

    education_score, health_score, civic_score, environmental_score, utilities_score

])

```

Note: This completes the scoring definition, but app integration for this sub-domain remains a Goal 5 data-pipeline task.

Sub-Domain 7: MOBILITY & TRANSPORTATION

App Questions: Q0351-Q0361 (Mobility)

Balance Sheet: Mobility Balance Sheet

Key Metrics & Scoring:

Vehicle adequacy

households_with_vehicle = count(Q0351 > 0)

vehicle_ratio = households_with_vehicle / total_households

vehicle_score = min(1.0, vehicle_ratio)

Unmet requirement

needs_vehicle = count(Q0357 == "Yes")

unmet_ratio = needs_vehicle / total_households

adequacy_score = 1.0 - unmet_ratio

Mobility_Score = average([vehicle_score, adequacy_score])

Domain 3: Livelihoods & Wellbeing

Sub-Domain 8: AGRICULTURE / FOOD

App Questions: Q0020-Q0202 (Food Production + Consumption)

Balance Sheet: Food Balance Sheet

Key Metrics & Scoring:

Food self-sufficiency

For each food type: self_grown / total_consumed

Aggregate across all food categories

total_self_produced = sum(Q0024, Q0053, Q0074, Q0087, Q0100, Q0113, ...) # all self-consumed

total_consumed = sum(Q0133, Q0139, Q0145, Q0151, Q0157, Q0163, ...) # all total qty consumed

self_sufficiency = total_self_produced / total_consumed if total_consumed > 0 else 0

```

if self_sufficiency >= 0.90:    food_score = 1.0

elif self_sufficiency >= 0.70:    food_score = linear(0.70, 0.7, 0.90, 1.0)

elif self_sufficiency >= 0.50:    food_score = linear(0.50, 0.5, 0.70, 0.7)

else:                            food_score = linear(0.0, 0.0, 0.50, 0.5)


# Soil health

stable_soils = count(Q0030 == "Stable") + count(Q0044 == "Stable")

total_soils = count(Q0030) + count(Q0044)

soil_ratio = stable_soils / total_soils if total_soils > 0 else NC

if soil_ratio >= 0.80:            soil_score = 1.0

elif soil_ratio >= 0.60:            soil_score = linear(0.60, 0.6, 0.80, 1.0)

else:                            soil_score = linear(0.0, 0.0, 0.60, 0.6)


# Organic farming ratio

organic_fertilizer = count(Q0032 in ["Organic Self Made", "Organic Purchased"])

total_fertilizer = count(Q0032)

organic_ratio = organic_fertilizer / total_fertilizer if total_fertilizer > 0 else NC

organic_score = organic_ratio


# Dietary diversity (from consumption section)

food_categories_consumed = count of non-zero categories in Q0131-Q0202

# Categories: Grains, Vegetables, Herbs, Legumes, Fruits, Dairy, Meat,

# Spices, Tea/Coffee, Oils, Processed, Alcohol/Tobacco

if categories >= 9:                diversity_score = 1.0

```

```

elif categories >= 6:          diversity_score = 0.7
elif categories >= 3:          diversity_score = 0.4
else:                          diversity_score = 0.1

# Storage adequacy

has_grain_storage = 1 if Q0121 != empty else 0
has_meat_storage = 1 if Q0125 != empty else 0
has_fruit_storage = 1 if Q0127 != empty else 0

storage_score = (has_grain_storage + has_meat_storage + has_fruit_storage) / 3

Food_Score = average([food_score, soil_score, organic_score,
                      diversity_score, storage_score])

```

Sub-Domain 9: BUSINESS & COMMERCIAL

App Questions: Q0399-Q0440 (Business & Commercial)

Balance Sheet: Business Balance Sheet

Key Metrics & Scoring:

```

# Business activity

households_with_business = count(Q0399 == "Yes")

business_ratio = households_with_business / total_households

# Business profitability

profitable = count(Q0411 > 0)

total_businesses = sum(Q0400)

profit_ratio = profitable / total_businesses if total_businesses > 0 else NC

# Employment generation

total_coworkers = sum(Q0408 + Q0409) # within + outside village

employment_score = min(1.0, total_coworkers / total_population * 10)

Business_Score = average([business_ratio, profit_ratio, employment_score])

```

Sub-Domain 10: INCOME & EXPENDITURE

App Questions: Q0212 (Income per member), Q0034/Q0048/Q0068/Q0092/Q0105/Q0118 (farm incomes), Q0411 (business income)

Balance Sheet: Income & Expenditure Balance Sheet

Key Metrics & Scoring:

Household income versus expenditure

Income sources: Q0212, farm incomes, business income

Expenditure: Q0035 (farm), Q0316/Q0320/Q0324/Q0328 (water), Q0340-Q0346 (energy),

Q0375-Q0395 (personal items), Q0069/Q0093/Q0106/Q0119 (livestock/fishery)

total_income = aggregate all income fields

total_expenditure = aggregate all expense fields

NEW: Minimum Material Needs Income Indicator

Sum of essential expenditures (water, energy, basic food, housing upkeep, healthcare)

Represents the absolute liability floor required to sustain life in the village.

minimum_material_needs = aggregate essential and baseline material expense fields

material_needs_met_households = count(household total_income >= minimum_material_needs)

material_needs_met_pct = material_needs_met_households / total_households

surplus_ratio = (total_income - total_expenditure) / total_income if total_income > 0 else 0

surplus_households = count(household surplus_ratio > 0)

surplus_pct = surplus_households / total_households

if surplus_pct >= 0.80: score = 1.0

elif surplus_pct >= 0.60: score = 0.7

elif surplus_pct >= 0.40: score = 0.4

else: score = 0.1

Income_Score = score

Sub-Domain 11: PERSONAL & HOUSEHOLD ITEMS

App Questions: Q0375-Q0398 (Personal & Household Items)

Balance Sheet: Personal & Household Care Balance Sheet

Key Metrics & Scoring:

Self-sufficiency (locally produced items)

```
locally_produced_count = count(Q0376=="Yes") + count(Q0380=="Yes") + count(Q0384=="Yes") + \
    count(Q0388=="Yes") + count(Q0392=="Yes") + count(Q0396=="Yes")
```

```
total_categories = 6 # personal care, cleaning, office, medicine, kitchen, other
```

```
self_sufficiency = locally_produced_count / total_categories
```

```
PersonalItems_Score = self_sufficiency
```

VILLAGE RESOURCE ADEQUACY SCORE

```
Village_Resource_Score = average([
    Land_Score,
    Water_Score,
    Forestry_Score,
    Energy_Score,
    Housing_Score,
    Community_Infra_Score, # NOTE: currently no app data for this
    Mobility_Score,
    Food_Score,
    Business_Score,
    Income_Score,
    PersonalItems_Score
])
```

PART C: INTEGRATION — VILLAGE SUSTAINABILITY SCORE

$$\text{Village_Sustainability} = (\text{Individual_Wellbeing} * 0.6) + (\text{Village_Resource} * 0.4)$$

Rationale: Individual wellbeing is weighted higher because it is the ultimate goal; resource adequacy is the means.

Score Interpretation:

Range	Level	Action
0.80 - 1.00	Excellent	Share best practices, maintain
0.60 - 0.80	Good	Enhance and strengthen
0.30 - 0.60	Moderate	Targeted intervention needed
0.00 - 0.30	Low	Immediate attention required

PART D: APP QUESTION → USE CASE MAPPING

Every question in `Data_Architecture_Map_yk.xlsx` serves one or more of these purposes:

- **Wellbeing (WB)** = Wellbeing Scoring (individual, feeds into 7 spheres)
- **Balance Sheet (BS)** = Balance Sheet (village resource scoring, feeds into 11 sub-domains)
- **ADMIN** = Administrative/registration (neither scoring nor balance sheet)
- **CONTEXT** = Provides context/qualitative data but doesn't feed a formula directly
- **DERIVE** = Can be used to derive/compute a scoring indicator

Registration (Q0001-Q0019)

Q_ID Range	Use Case	Details
Q0001-Q0019	ADMIN	Household and moderator registration. Sets up country, location, identity.

Food Production (Q0020-Q0130)

Q_ID Range	Sub-Category	Use Case	Feeds Into
Q0020	Cultivation display	Balance Sheet (BS)	Land Balance Sheet (land allocation)
Q0021-Q0029	Crop type, area, output, distribution	Balance Sheet (BS)	Food Balance Sheet (production side)
Q0030-Q0031	Soil Health	Balance Sheet (BS) + Wellbeing (WB)(derive)	Food Score (soil_score), Land Score

Q_ID Range	Sub-Category	Use Case	Feeds Into
Q0032-Q0033	Fertiliser/Pesticide type	Balance Sheet (BS) + Wellbeing (WB)(derive)	Food Score (organic_score)
Q0034-Q0035	Income/Expenditure from crops	Balance Sheet (BS)	Income Balance Sheet, Food Balance Sheet
Q0036	Processing method	CONTEXT	
Q0037	Yield per unit	Balance Sheet (BS)	Food Balance Sheet
Q0038-Q0039	Planting/Harvesting month	Balance Sheet (BS)	Farming calendar, seasonal analysis
Q0040-Q0060	Trees, Shrubs & Grasslands (same pattern)	Balance Sheet (BS)	Food Balance Sheet
Q0061-Q0080	Poultry/Livestock (same pattern)	Balance Sheet (BS)	Food Balance Sheet
Q0081-Q0108	Fishery — Pond + River (same pattern)	Balance Sheet (BS)	Food Balance Sheet
Q0109-Q0120	Hunting (same pattern)	Balance Sheet (BS)	Food Balance Sheet
Q0121-Q0129	Storage	Balance Sheet (BS) + Wellbeing (WB)(derive)	Food Score (storage_score)
Q0130	Selling Channels	Balance Sheet (BS)	Income Balance Sheet

Food Consumption (Q0131-Q0202)

Q_ID Range	Sub-Category	Use Case	Feeds Into
Q0131-Q0202	12 food categories × (types, qty, market/neighbor/self-grown)	Balance Sheet (BS) + Wellbeing (WB)(derive)	Food Balance Sheet (consumption side), Food Score (self_sufficiency, diversity), Physical Wellbeing (dietary balance proxy)

Demographic Information (Q0203-Q0265)

Q_ID Range	Sub-Category	Use Case	Feeds Into
Q0203	Family size	Balance Sheet (BS)	All balance sheets (population data)
Q0204	Marital status	CONTEXT	
Q0205	Diet type	Wellbeing (WB)(derive)	Physical: P2.3 dietary balance (partial)
Q0206	Height	Wellbeing (WB)	Physical: P1.1 BMI
Q0207	Weight	Wellbeing (WB)	Physical: P1.1 BMI
Q0208-Q0210	Languages	CONTEXT	Community profile
Q0211	Occupation	Balance Sheet (BS) + CONTEXT	Income/Business Balance Sheet
Q0212	Yearly Income	Wellbeing (WB) + Balance Sheet (BS)	Occupational: O1.1 Annual Livelihood Income Level + Income Balance Sheet

Q_ID Range	Sub-Category	Use Case	Feeds Into
Q0213	Bank Account	Wellbeing (WB)(derive)	Occupational: O2.6 Financial & Enterprise Literacy (partial)
Q0214	Savings Y/N	Wellbeing (WB)	Occupational: O1.2 Savings & Investment Status
Q0215	Savings Amount	Balance Sheet (BS)	Income Balance Sheet
Q0216	Chronic Diseases	Wellbeing (WB)(derive)	Physical: P3.4 Chronic Disease (Tier 3)
Q0217	Motor Disability	CONTEXT	Community needs assessment
Q0218	Overall Well-being	Wellbeing (WB)	Emotional: E1.1 Current State
Q0219	Communication of Feelings	Wellbeing (WB)	Emotional: E2.3 Connection (proxy)
Q0220	Support System	Wellbeing (WB)	Emotional: E1.2 Support System
Q0221	Stress and Resilience	Wellbeing (WB)	Emotional: E2.1 Stress Frequency
Q0222	Empathy	Wellbeing (WB)	Social: S1.2 Empathy Level
Q0223	Habits	CONTEXT	Physical wellbeing context
Q0224	Education Status	Wellbeing (WB)	Intellectual: I1.1 Education Level
Q0225	Education Seeking	Wellbeing (WB)	Intellectual: I1.2 Seeking Education
Q0226-Q0230	Current Skillset	Balance Sheet (BS) + CONTEXT	Village skill inventory
Q0231	Cultural Hobbies	Wellbeing (WB)	Social: S1.1 Cultural Participation
Q0232-Q0235	Other Hobbies	CONTEXT	Community profile
Q0236-Q0242	Skills Seeking to Learn	Wellbeing (WB)	Intellectual: I1.3 Seeking Skills
Q0243-Q0250	Aspirations	CONTEXT	Village planning, needs assessment
Q0251	Spiritual Aspirations	Wellbeing (WB)(partial)	Spiritual: SP2.1 Meditation (aspiration only, not practice)
Q0252-Q0259	Unfulfilled Needs	CONTEXT	Gap analysis, intervention priorities
Q0260-Q0265	Wishes	CONTEXT	Community vision

Landholding & Usage (Q0266-Q0288)

Q_ID Range	Use Case	Feeds Into	
Q0266-Q0284	Land inventory (count, location, area, use, type)	Balance Sheet (BS) + Wellbeing (WB)(derive)	Land Balance Sheet, Occupational: O2.5 Asset Ownership
Q0285-Q0288	Land requirements	Balance Sheet (BS)	Land Balance Sheet (unmet needs)

Housing (Q0289-Q0312)

Q_ID Range	Use Case	Feeds Into	
Q0289-Q0300	House inventory (count, type, area, floors, materials, photos)	Balance Sheet (BS) + Wellbeing (WB)(derive)	Housing Balance Sheet, Occupational: O2.5 Asset Ownership, Housing Score
Q0301-Q0303	Amenities, equipment, furnishing	Balance Sheet (BS)	Housing Balance Sheet
Q0304-Q0307	Renovation/Expansion needs	Balance Sheet (BS)	Housing Balance Sheet (unmet needs)
Q0308-Q0312	New housing requirements	Balance Sheet (BS)	Housing Balance Sheet (deficit)

Water (Q0313-Q0337)

Q_ID Range	Use Case	Feeds Into	
Q0313-Q0316	Cooking/Drinking water (consumption, source, quality, expense)	Wellbeing (WB) + Balance Sheet (BS)	Environmental: EN1.1 Water Quality + Water Balance Sheet
Q0317-Q0320	Sanitation water	Wellbeing (WB) + Balance Sheet (BS)	Environmental: EN1.2 + Water Balance Sheet
Q0321-Q0324	Cleaning water	Balance Sheet (BS)	Water Balance Sheet
Q0325-Q0328	Irrigation water	Balance Sheet (BS)	Water Balance Sheet
Q0329	Other water purposes	Balance Sheet (BS)	Water Balance Sheet
Q0330-Q0333	Harvest & Wastewater	Balance Sheet (BS)	Water Balance Sheet
Q0334-Q0337	General (meter, scarcity)	Balance Sheet (BS)	Water Balance Sheet, Water Score

Energy & Fuel (Q0338-Q0350)

Q_ID Range	Use Case	Feeds Into	
Q0338	Grid connection	Balance Sheet (BS)	Energy Balance Sheet, Energy Score
Q0339-Q0346	Electricity/Fuel consumption & expense	Balance Sheet (BS)	Energy Balance Sheet
Q0347	Other fuel sources	Balance Sheet (BS)	Energy Balance Sheet
Q0348	Microgrid installed	Balance Sheet (BS)	Energy Score (renewable)
Q0349-Q0350	Village renewable energy	Balance Sheet (BS)	Energy Score

Mobility (Q0351-Q0361)

Q_ID Range	Use Case	Feeds Into	
Q0351-Q0356	Vehicle inventory	Balance Sheet (BS) + Wellbeing (WB)(derive)	Mobility Balance Sheet, Occupational: O2.5 Asset Ownership

Q_ID Range	Use Case	Feeds Into	
Q0357-Q0361	Vehicle requirements	Balance Sheet (BS)	Mobility Balance Sheet (unmet needs)

Forestry & Timber (Q0362-Q0374)

Q_ID Range	Use Case	Feeds Into	
Q0362-Q0368	Forest usage (area, harvest, produce)	Balance Sheet (BS)	Forestry Balance Sheet, Land Score
Q0369-Q0374	Timber/produce requirements	Balance Sheet (BS)	Forestry Balance Sheet (demand side)

Personal & Household Items (Q0375-Q0398)

Q_ID Range	Use Case	Feeds Into	
Q0375-Q0398	6 categories × (expense, locally produced Y/N, type, qty)	Balance Sheet (BS) + Wellbeing (WB)(derive)	Personal Items Balance Sheet, Personal Items Score

Business & Commercial (Q0399-Q0440)

Q_ID Range	Use Case	Feeds Into	
Q0399-Q0419	Business inventory (type, income, expenses, resources)	Balance Sheet (BS) + Wellbeing (WB)(derive)	Business Balance Sheet, Occupational: O2.5 Asset Ownership, Income Balance Sheet (BS)
Q0420-Q0431	Business requirements	Balance Sheet (BS)	Business Balance Sheet (unmet needs)
Q0432-Q0436	New business plans	CONTEXT + Balance Sheet (BS)	Business Balance Sheet
Q0437-Q0440	Village-level establishments	Balance Sheet (BS)	Business Balance Sheet

PART E: GAPS — QUESTIONS MISSING FROM THE APP

Critical Gap: Wellbeing Survey Questions Not in App

The following indicators have NO corresponding question in the current 445-question app. These need to be added as a new "Individual Wellbeing" section:

MUST ADD (Tier 1+2 — needed for Malaysia 2026):

Indicator	Question Needed	Sphere
P2.1	Sleep hours + quality	Physical
P2.2	Exercise frequency + duration	Physical
P2.4	Daily energy levels	Physical

Indicator	Question Needed	Sphere
P2.5	Dental hygiene (brushing + problems)	Physical
O2.1	Livelihood adequacy (subjective)	Occupational
O2.2	Occupational security feeling	Occupational
O2.3	Emergency fund duration	Occupational
O2.4	Debt burden	Occupational
E2.2	Anger/temperament frequency	Emotional
E2.4	Life satisfaction	Emotional
S2.2	Neighbor relations quality	Social
S2.3	Community contribution level	Social
S2.4	Sense of belonging	Social
S2.5	Social conflicts past year	Social
EN2.1	Air quality perception	Environmental
EN2.2	Green space access + distance	Environmental
EN2.3	Neighborhood cleanliness	Environmental
EN2.4	Garbage management type	Environmental
EN2.5	Land/space adequacy (subjective)	Environmental
EN2.6	Food quality access	Environmental
I2.1	Value of education/learning	Intellectual
I2.2	Learning engagement frequency	Intellectual
I2.3	Information access	Intellectual
SP1.1	Spiritual self-evaluation	Spiritual
SP1.2	Ethical/moral consideration frequency	Spiritual
SP2.1	Meditation/prayer frequency	Spiritual
SP2.2	Silence/reflection frequency	Spiritual
SP2.3	Sense of purpose	Spiritual
SP2.4	Gratitude practice frequency	Spiritual
SP2.5	Mindfulness in daily actions	Spiritual

Total new questions needed: ~30 questions (some are 2-part)

Other Gaps:

Gap	Description	Priority
Community Infrastructure	No questions about village-level facilities (schools, hospitals, markets, etc.)	HIGH — needed for Sub-Domain 6
Income & Expenditure section	No dedicated section in app — currently scattered across other sections	MEDIUM — can be computed from existing data

Gap	Description	Priority
Village-level totals	App collects household data but some metrics need village-level totals (total area, total population from census)	MEDIUM — may come from admin records

SUMMARY: Data Source for Each Scoring Component

Scoring Component	Data Source	Status
Physical Wellbeing	App (Q0206-Q0207) + New wellbeing questions	PARTIAL — BMI only
Occupational Wellbeing	App (Q0212, Q0214-Q0215) + New questions	PARTIAL — currently operationalized mainly through income, savings, assets, debt, and livelihood resilience
Emotional Wellbeing	App (Q0218-Q0221) + New questions	GOOD — 3 of 6 indicators covered
Social Wellbeing	App (Q0222, Q0231) + New questions	PARTIAL — 2 of 7 indicators
Environmental Wellbeing	App (Q0315, Q0319) + New questions	PARTIAL — 2 of 8 indicators
Intellectual Wellbeing	App (Q0224-Q0225, Q0236-Q0242) + New questions	GOOD — 3 of 6 indicators covered
Spiritual Wellbeing	New questions only	MISSING — 0 of 7 indicators
Land Score	App (Q0266-Q0288, Q0362)	GOOD
Water Score	App (Q0313-Q0337)	GOOD
Forestry Score	App (Q0362-Q0374)	GOOD
Energy Score	App (Q0338-Q0350)	GOOD
Housing Score	App (Q0289-Q0312)	GOOD
Community Infra Score	NOT IN APP	MISSING
Mobility Score	App (Q0351-Q0361)	GOOD
Food Score	App (Q0020-Q0202)	EXCELLENT
Business Score	App (Q0399-Q0440)	GOOD
Income Score	Computed from scattered fields	MODERATE
Personal Items Score	App (Q0375-Q0398)	GOOD

FIXES APPLIED IN THIS DOCUMENT

Issue	Old Behavior	New Behavior	Rationale
BMI scoring	Binary: 1 if normal, 0 otherwise	Graduated: 1.0/0.6/0.3/0.0 based on severity	Borderline cases shouldn't score zero
Occupational aggregation	<code>income * savings</code> (multiplication)	<code>average(income, savings)</code>	Multiplication is too punitive; person with high income but no savings scores 0

Issue	Old Behavior	New Behavior	Rationale
Savings "No" score	0.5	0.0	Having no savings should not score neutral
Emotional state options	3 options	4 options (added "Facing Challenges" = 0.25)	App has 4 options
Support system	Binary (Y/N)	Graduated by count of support types	App allows multi-select
Water quality options	3 options	4 options (added "Slightly Contaminated" = 0.25)	App has 4 options
Education levels	4 options	12 options mapped to same scale	App has more granular options
All sphere aggregation	Mixed (some multiply, some average)	Always average	Consistency

This document is the single source of truth for wellbeing scoring. For historical reference, see `IMPLEMENTATION_SPECIFICATION.md` (Bhutan pilot logic) and `FRAMEWORK_EXTENSION_ANALYSIS.md` (extension analysis).